Statistical Issues in Multicenter Randomized Clinical Trials

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Abstract

Multicenter randomized controlled trials require a choice among ignoring, fixing, or randomizing site effects in the primary analysis, and this choice interacts with stratified-randomization adjustment, treatment-by-site interaction, and site-size balance. We report a factorial Monte Carlo simulation, structured under the ADEMP framework of Morris, White, and Crowther (2019), that jointly varies the number of sites, site-level variance, treatment-by-site interaction, and site-size balance, comparing OLS ignoring site, OLS with fixed site effects, and a linear mixed model with random site effects. In addition to bias, empirical and model-based standard error, power, and coverage under a true alternative treatment effect, we report empirical Type I error under a true null effect, and singular/boundary-fit rates for the random-effects model, both previously unreported in this simulation design. The random-effects model attains the best combination of power and coverage across most conditions; ignoring site produces conservative (not anti-conservative) inference under stratified randomization, consistent with prior work; and all three methods control Type I error near the nominal level in the null-hypothesis scenarios simulated. For the treatment-by-site-interaction scenarios, where the random-intercept model is misspecified, we additionally fit and compare a random-slopes model, finding consistently higher coverage but a substantially higher singular-fit rate, a trade-off not previously quantified in this simulation design. Every point estimate is reported with its Morris et al. Table 6 Monte Carlo standard error. Code and a full ADEMP audit trail accompany the simulation.

Contents

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1 Introduction

Multicenter randomized clinical trials (RCTs) represent the standard design for confirmatory evaluation of therapeutic interventions. By enrolling participants across multiple clinical sites, these trials achieve practical recruitment targets while broadening the population base for generalization of findings[1,2]. However, the multicenter structure introduces several analytic complexities that, if mishandled, can distort inference about treatment effects. We address four interrelated statistical issues that arise in the design and analysis of multicenter RCTs with continuous outcomes, and we present simulation evidence bearing on each in turn.

1.1 Modeling site effects

The first question concerns how site (or center) effects should be incorporated into the primary analysis. Three broad strategies exist: ignoring site differences entirely, treating sites as fixed effects, and treating sites as random effects drawn from a population of potential sites. Each strategy carries distinct assumptions and implications for the validity and efficiency of treatment effect estimation[3,4].

Localio and colleagues provided an influential overview of the problem, identifying three analytic challenges inherent to multicenter data: within-center correlation of outcomes, confounding by center, and effect modification across centers[2]. The choice between fixed and random effects models has been debated extensively. Fixed effects models condition on the observed sites and make no distributional assumptions about site effects, but they consume degrees of freedom rapidly when the number of sites is large relative to total sample size[5]. Random effects models treat site effects as realizations from a normal distribution, estimating only a variance component rather than individual site parameters, and thereby preserve statistical efficiency[6]. The ICH E9 guideline recommends that mixed models are 'especially relevant when the number of sites is large'[1].

Chu and colleagues conducted a comprehensive simulation comparing six analytic approaches for continuous outcomes across varying numbers of centers, center sizes, and intraclass correlation coefficients (ICCs)[7]. All methods yielded unbiased point estimates, but ignoring centers inflated the standard error and reduced power when within-center correlation was present. The mixed-effects model attained nominal coverage and near-optimal power across nearly all configurations. Generalized estimating equations (GEE) underestimated the standard error when the number of centers was small.

Kahan and Morris extended these findings, demonstrating that random center effects performed at least as well as fixed effects in all scenarios examined, and substantially outperformed them when the number of patients per center was small[8]. Islam and Bangdiwala confirmed these patterns in a recent replication study with both continuous and binary outcomes, reinforcing the recommendation for random effects models as a default analytic strategy[9].

1.2 What does "random" actually mean here?

Before going further, it is worth being precise about the word "random," because this word carries most of the weight in the discussion above, and because it can mean two different things. We spell out both meanings in plain terms, with the underlying math shown alongside, since the two meanings lead to two different arguments later in this Introduction.

1.2.1 A hospital-specific shift, fixed or random

Picture a trial run at ten hospitals. Each hospital has its own typical outcome level, for reasons that have nothing to do with the treatment being tested: a different mix of patients, different routine practices, different equipment. Call that hospital-specific shift α_i, for hospital i. A simple model for one patient's outcome, written Y_{ij} for patient j at hospital i, looks like this:

$$Y_{ij} = \alpha_i + \delta\, T_{ij} + \epsilon_{ij}$$

Here T_{ij} is 1 if that patient received the active treatment and 0 if not, δ is the treatment effect the trial wants to estimate, and ϵ_{ij} is ordinary patient-to-patient variation that has nothing to do with which hospital a patient attended.

The open question is what kind of quantity α_i is.

There are two options. The first treats each hospital's shift as an ordinary, fixed number, no different in kind from a patient's age or sex: something to adjust for, one number per hospital. Ten hospitals means ten fixed numbers to estimate.

The second option treats α_i as random, drawn from a normal distribution: $\alpha_i \sim N(0, \sigma^2_\alpha)$. Under this option, the one unknown quantity of real interest is σ^2_α: how much do hospitals typically differ from one another? Rather than estimating ten separate numbers, the analysis estimates a single number describing the *spread* of hospital effects.

This is not a small technical difference. It changes what the analysis is entitled to claim. Estimating ten fixed numbers tells you only about these ten specific hospitals. Estimating one spread, σ_α^2, is a broader claim: it says something about hospital-to-hospital variation in general, of the kind you would expect if the trial had, by chance, enrolled a different set of ten hospitals instead.

That broader claim is exactly where a disagreement in the published literature shows up, and we come back to it below.

1.2.2 Why the simulation needs a genuinely random site effect

This paper is a computer simulation. It generates many pretend trials, over and over, and checks how well each analysis strategy (ignore site, treat site as fixed, treat site as random) recovers the true treatment effect and reports honest uncertainty around it. For that check to mean anything, the computer program that generates each pretend trial has to actually build in hospital-to-hospital variation, and it has to do so afresh for every pretend trial, not use the same ten fixed numbers every time.

That is what drawing $\alpha_i \sim N(0, \sigma_\alpha^2)$ inside the simulation accomplishes: every pretend trial gets its own new set of hospital shifts, all coming from a distribution with the same spread, σ_α^2. Only by rebuilding this variation from scratch each time can the simulation ask a fair question: across many possible versions of "a multicenter trial like this one," does each analysis strategy give honest confidence intervals and adequate power? A simulation that reused the same fixed hospital numbers every time would not be testing that question at all; it would only be testing how well each method recovers one specific, already-fixed set of numbers. The later sections on treatment-by-site interaction and on imbalanced site sizes work the same way: each also needs a freshly drawn random ingredient, every pretend trial, for the same reason.

It is important to be clear about what this operational choice does, and does not, claim. It is a requirement of building a fair test, not a statement about how real hospitals were chosen for a real trial. That second, separate question is the subject of the next section.

1.2.3 Two different reasons statisticians give for treating site as random

Everything so far concerns how the simulation is built. A separate question is why, in a real trial with real hospitals, an analyst should choose to treat site as random rather than fixed when fitting the actual statistical model. Published statisticians do not fully agree on the reason, even where they agree on the recommendation, and a statistically literate reader deserves to know both arguments.

The first argument says the random-effects model literally means what it appears to mean: it treats the hospitals in the trial as if they had been drawn at random from a much larger population of possible hospitals. Falissard and Chavance make this point directly: the random-effects interpretation is only truly justified when the hospitals really were chosen at random from a wider population[10]. In practice, that is almost never how hospitals are chosen. Hospitals join a trial because they can recruit patients quickly, because an investigator has a relationship with the study team, or because of funding and logistics, not by a lottery over all possible hospitals[3].

The second argument does not depend on how the hospitals were chosen at all. Kahan and Morris justify adjusting for site on the design of the trial itself: patients were randomly split between treatment and control separately within each hospital, so the analysis should reflect that split, no matter how the hospitals themselves came to be in the trial[8,11]. Edgar, Roberts, and Sharples reach the same conclusion by re-analyzing a real trial run across 271 hospitals in 40 countries: they recommend a random site effect for practical reasons, better power and less bias, again without any claim about hospitals being a random sample of a larger population[12].

These two arguments do not agree on *why* a random site effect is the right choice, but they agree on *what to do*: prefer a random effect for site once there are more than a handful of hospitals. The recommendation in this paper rests on that agreement, so it stands regardless of which argument a reader finds more convincing. The simulation itself follows the second, design-based tradition in spirit: it builds hospital-to-hospital variation into every pretend trial as an operational necessity for testing the analysis strategies fairly (see above), without needing or claiming that real hospitals are ever a literal random sample of all possible hospitals.

1.3 Failure to adjust for stratification variables

The second issue concerns the consequences of ignoring stratification variables, including site, in the analysis. When randomization is stratified by center (as is standard in multicenter trials), failing to adjust for the stratification factor in the analysis leads to biased inference. Kahan and Morris demonstrated through simulation that an unadjusted analysis of a trial randomized with stratified blocks produces standard errors that are biased upward, confidence intervals that are too wide, type I error rates below the nominal level, and a loss of statistical power[11]. This paradoxical conservatism arises because the stratified randomization induces negative correlation between treatment groups within strata; ignoring this correlation inflates variance estimates.

Kahan and Morris extended this work to trials with multiple stratification factors, finding that adjusting for the main effects of all stratification variables was generally sufficient without requiring adjustment for all cross-classified strata[13]. These findings align with the ICH E9 recommendation that 'the statistical model used for analysis should reflect the restriction imposed by the randomisation procedure'[1], and with current regulatory guidance on covariate adjustment more broadly[14].

1.4 Treatment-by-site interaction

The third concern is heterogeneity of the treatment effect across sites. Even under a common protocol, eligible participants, site practices, and investigator experience may vary, producing genuine treatment-by-site interaction[15]. Senn argued that some degree of treatment-by-center interaction is inevitable in practice and that a rational approach to planning multicenter trials should anticipate it[3].

The ICH E9 guideline recommends first fitting a model without the interaction term to estimate the main treatment effect, and then examining heterogeneity as a secondary analysis. However, the power to detect treatment-by-site interaction is typically low, so a non-significant interaction test does not establish homogeneity[16]. Jones and colleagues compared fixed-effects weighted estimators, unweighted estimators, and random effects estimators in the presence of varying degrees of treatment-by-center interaction[4]. The fixed effects weighted estimator performed well across a range of interaction magnitudes, while unweighted estimators could be inefficient when site sizes differed markedly.

When treatment-by-site interaction is modeled as random (i.e., the treatment slope varies across sites), the mixed model estimates both the average treatment effect and the variance of treatment effects across sites, providing a natural summary of heterogeneity[17].

1.5 Imbalanced site sizes

The fourth issue, closely related to the preceding three, is whether imbalance in site sizes affects inference. Multicenter trials frequently exhibit highly unequal enrollment across sites, with a few large sites contributing a disproportionate share of participants. Senn noted that a rational approach to planning multicenter trials leads naturally to unequal site sizes, because sites with higher recruitment capacity will enroll more patients within the trial's fixed time window[3].

Ruvuna introduced a coefficient of imbalance to quantify the power loss attributable to unequal center sizes and showed that substantial imbalance can reduce power when an unweighted (Type III) analysis is used[18]. Vierron and Giraudeau formalized this through a design effect framework, demonstrating that the efficiency loss depends on both the ICC and a statistic quantifying the heterogeneity of group distributions across centers[19]. When the ICC is positive, balanced allocation is optimal, but moderate imbalance imposes only a modest efficiency penalty. The key risk emerges when site size is informative (that is, when larger sites systematically differ from smaller sites in patient characteristics or treatment delivery), potentially introducing bias into the treatment effect estimate[15,20].

1.6 Present study

The existing literature provides clear theoretical guidance but limited integrated simulation evidence spanning all four issues simultaneously. We present a Monte Carlo simulation that jointly varies the analytic method (ignore site, fixed site effects, random site effects), the presence or absence of treatment-by-site interaction, balanced versus imbalanced site sizes, and the magnitude of site-level variance. The simulation evaluates bias, empirical standard error, model-based standard error, power, and coverage probability of 95% confidence intervals for the average treatment effect across all factorial combinations of these design features.

2 Methods

2.1 ADEMP structure

Following Morris, White, and Crowther[21], the simulation is reported under the ADEMP framework.

- **Aims.** Compare three primary analytic strategies for handling site heterogeneity in multicenter RCTs (ignore site, fixed site effects, random site effects) across scenarios varying the number of sites, site variance, treatment-by-site interaction, and site-size balance; separately assess whether these strategies control the Type I error rate under a true null treatment effect; and, for the treatment-by-site-interaction scenarios specifically, compare the random-intercept specification against a fourth, correctly specified random-slopes method.
- **Data-generating mechanism.** Detailed in the next subsection.
- **Estimand.** The overall treatment effect δ on the outcome scale, in the simulation-performance sense of Morris et al. (a data-generating parameter with a known true value), not in the ICH E9(R1) sense of a fully specified population/variable/ intercurrent-event/summary-measure estimand[22].
- **Methods.** OLS ignoring site, OLS with fixed site effects, a linear mixed model with random site effects, and (fit on every replicate for a paired comparison, but reported only for the treatment-by-site-interaction scenarios) a linear mixed model with random site effects and a random treatment slope, all via `lme4::lmer`. All four methods are tested and interval-estimated using a t-reference distribution: the OLS residual degrees of freedom for the ignore-site and fixed-site methods, and N minus the number of fixed-effect parameters for the two mixed-model methods. The latter is a naive approximation, not a Kenward-Roger or Satterthwaite denominator-df correction (see Future research); it is used consistently for both the p-value and the 95% CI half-width for both mixed-model methods, replacing an earlier implementation in which the p-value used this t-approximation but the CI used a fixed $z = 1.96$ half-width regardless of sample size or method.
- **Performance measures.** Bias, empirical SE, MSE, mean model SE, power at $\alpha = 0.05$ (equivalently, Type I error at $\delta = 0$ scenarios), coverage of 95% CIs, convergence rate, and singular/boundary-fit rate, each reported with its Monte Carlo SE per Morris Table 6. Non-convergence (a hard error from `lmer`) and singular/boundary fits (a successful fit at the edge of the random-effects covariance parameter space, which `lme4` flags as a warning rather than an error) are tracked as two distinct first-class outcomes and reported per scenario and method, rather than treating a fit that raised any warning as simply non-convergent or, conversely, absorbing a boundary fit silently into "converged" with no record that it was a boundary solution.

2.2 Data generating model

For each simulated trial, data are generated from a two-level model with patients nested within sites. Let Y_{ij} denote the continuous outcome for patient j at site i:

$$Y_{ij} = \alpha_i + (\delta + \gamma_i)T_{ij} + \epsilon_{ij}$$

where $\alpha_i \sim N(0, \sigma_\alpha^2)$ is the random site intercept, $\delta = 0.30$ is the true average treatment effect, $\gamma_i \sim N(0, \sigma_\gamma^2)$ is the random treatment-by-site interaction, $T_{ij} \in \{0, 1\}$ is the treatment indicator (balanced within each site via permuted blocks), and $\epsilon_{ij} \sim N(0, \sigma_e^2 = 1)$ is the residual error.

2.3 Parameter values

The simulation crosses the following factors:

- **Number of sites:** $K = 10$ (few) vs $K = 30$ (many)
- **Site-level variance:** $\sigma_\alpha^2 \in \{0, 0.10, 0.50\}$ (none, moderate, large)
- **Treatment-by-site interaction:** $\sigma_\gamma^2 \in \{0, 0.04\}$ (none vs present)
- **Site size balance:** balanced ($n_i = 20$ for all i) vs imbalanced (site sizes drawn from a $\text{Gamma}(0.8, 0.8/20)$ distribution, yielding a coefficient of variation ≈ 1.1)

The true treatment effect is $\delta = 0.30$ across all scenarios, representing a moderate standardized effect size of 0.30 standard deviations.

2.4 Analytic methods

Each simulated dataset is analyzed by three primary methods, plus a fourth fit on every replicate for a paired comparison but reported only for the treatment-by-site-interaction scenarios (Results, "Random slopes under treatment-by-site interaction"):

1. **Ignore site:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + e_{ij}$ (ordinary least squares, no site term)
2. **Fixed site effects:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + \sum_{k=2}^{K} \beta_k I(i = k) + e_{ij}$ (site indicators as fixed effects)
3. **Random site effects:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_i + e_{ij}$ where $u_i \sim N(0, \sigma_u^2)$ (linear mixed model via REML, fit with `lme4`)
4. **Random slopes:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_i + v_i T_{ij} + e_{ij}$ where (u_i, v_i) jointly follow a bivariate normal distribution (random intercept and random treatment slope per site, via REML). This is the correctly specified model under the treatment-by-site-interaction data-generating mechanism (Data generating model, γ_i), unlike Method 3, which omits the random slope and is misspecified whenever $\sigma_\gamma^2 > 0$.

2.5 Simulation procedure

Scenario 1/24: K10_noVar_bal … Scenario 2/24: K10_modVar_bal … Scenario 3/24: K10_hiVar_bal … Scenario 4/24: K10_modVar_imbal … Scenario 5/24: K10_hiVar_imbal … Scenario 6/24: K10_modVar_int … Scenario 7/24: K10_hiVar_int … Scenario 8/24: K10_hiVar_int_imb … Scenario 9/24: K30_noVar_bal … Scenario 10/24: K30_modVar_bal … Scenario 11/24: K30_hiVar_bal … Scenario 12/24: K30_modVar_imbal … Scenario 13/24: K30_hiVar_imbal … Scenario 14/24: K30_modVar_int … Scenario 15/24: K30_hiVar_int

… Scenario 16/24: K30_hiVar_int_imb … Scenario 17/24: H0_K10_noVar_bal … Scenario 18/24: H0_K10_modVar_bal … Scenario 19/24: H0_K10_hiVar_bal … Scenario 20/24: H0_K10_hiVar_imbal … Scenario 21/24: H0_K30_noVar_bal … Scenario 22/24: H0_K30_modVar_bal … Scenario 23/24: H0_K30_hiVar_bal … Scenario 24/24: H0_K30_hiVar_imbal …

Each scenario is replicated $R = 1{,}500$ times. The Monte Carlo SE of an estimated coverage or Type I error probability p is MCSE $= \sqrt{p(1-p)/R}$ (Morris, White, and Crowther 2019, Table 6); evaluated at the nominal $p = 0.95$ (worst case among the coverage targets considered here), $R \geq 1{,}320$ is required for MCSE ≤ 0.6 percentage points, and $R = 1{,}500$ gives MCSE ≈ 0.56 pp, satisfying the target with a margin. Treatment is assigned via balanced allocation within each site (equal numbers of treatment and control participants, or within ± 1 when n_i is odd). All random number generation uses a fixed seed (`set.seed(20260310)`) and the `L'Ecuyer-CMRG` RNG for reproducibility. The `sim-run` chunk's cache is keyed on the MD5 hash of `analysis/scripts/sim_multicenter.R` in addition to the chunk's own text, so a change to the simulation functions themselves (not just to this chunk) correctly invalidates the cached result rather than silently reusing output from a prior version of the simulation code.

3 Results

3.1 Headline findings

We observe that across 16 alternative- hypothesis scenarios ($\delta = 0.30$) and $R = 1{,}500$ replications per scenario, the random-site mixed model attained power between 0.54 and 0.97, with coverage of the 95% confidence interval ranging from 0.895 to 0.960 and absolute bias never exceeding 0.011 (Morris Table 6 Monte Carlo SEs are given in parentheses next to the power and coverage point estimates in Table 1). The fixed-site estimator closely tracked the random-site estimator. Ignoring site entirely produced overcoverage (up to 0.984) and correspondingly lower power (as low as 0.40) in scenarios with non-trivial between-site variance. The lowest convergence rate (hard `lmer` errors only) observed across any scenario or method was 1.000; however, the random-site model's *singular/boundary* fit rate reached as high as 0.557 in the low- and no-site-variance scenarios, where the true random-intercept variance is at or near the parameter-space boundary. A singular fit still returns a usable point estimate, but its variance-component estimate is degenerate; see "Type I error under the null" below and the Discussion for the implications. Under the null-hypothesis scenarios ($\delta = 0$; Table 2), the empirical Type I error rate for the random-site method ranged from 0.041 to 0.068, and for the ignore-site method from 0.014 to 0.047, against a nominal $\alpha = 0.05$.

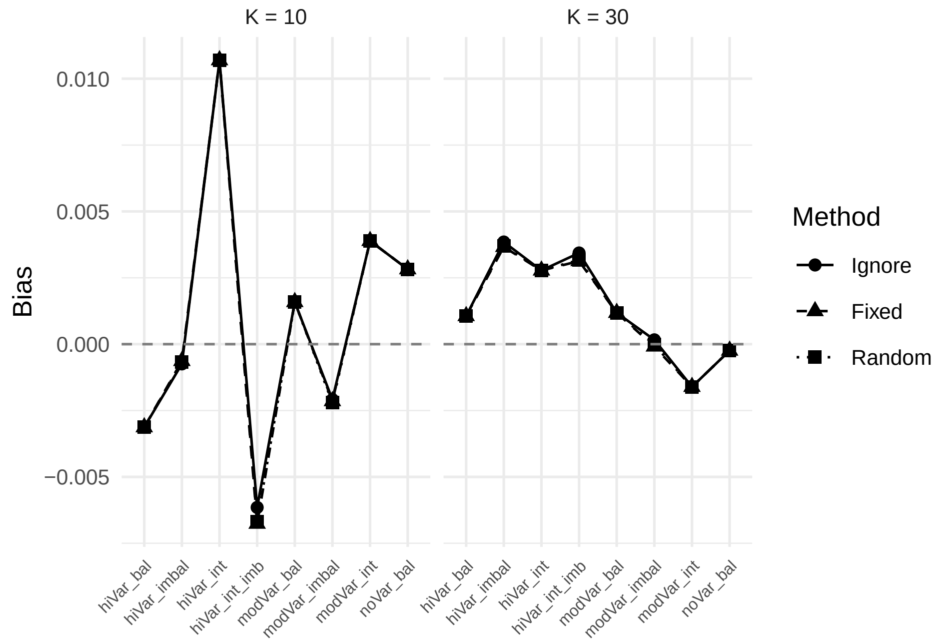

Figure 1: Bias of the treatment effect estimate across scenarios and analytic methods. Dashed line at zero indicates no bias.

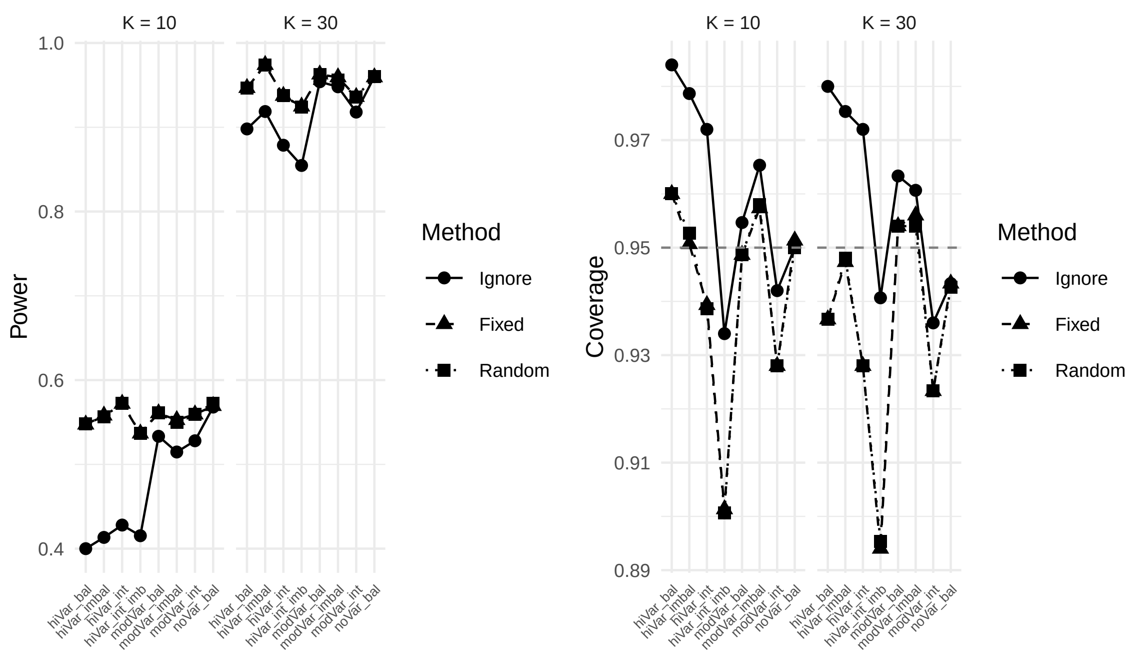

Figure 2: Power (left) and coverage probability (right) across scenarios. Dashed line indicates the 95% coverage target (right panel).

Table 1: Simulation results under the alternative hypothesis ($\delta = 0.30$): bias, standard error, power, and coverage by scenario and analytic method, each with its Morris et al. (2019) Table 6 Monte Carlo SE in parentheses where applicable. R = 1500 replications per scenario. 'Conv.' is the rate of successful `lmer` fits (errors only); 'Sing.' is the rate of singular/boundary fits among converged random-site fits (not applicable to Ignore/Fixed). Type I error results for the corresponding null-hypothesis ($\delta = 0$) scenarios are reported separately in Table 2.

Scenario	Method	Bias	Emp. SE	Model SE	Power (MCSE)	Coverage (MCSE)	Conv.	Sing.
K10 hiVar bal	Fixed	-0.003	0.140	0.141	0.547 (0.013)	0.960 (0.005)	100.0%	0.0%
K10 hiVar bal	Ignore	-0.003	0.140	0.169	0.400 (0.013)	0.984 (0.003)	100.0%	0.0%
K10 hiVar bal	Random	-0.003	0.140	0.141	0.548 (0.013)	0.960 (0.005)	100.0%	0.0%
K10 hiVar imbal	Fixed	-0.001	0.141	0.141	0.558 (0.013)	0.951 (0.006)	100.0%	0.0%
K10 hiVar imbal	Ignore	-0.001	0.142	0.167	0.413 (0.013)	0.979 (0.004)	100.0%	0.0%
K10 hiVar imbal	Random	-0.001	0.141	0.141	0.557 (0.013)	0.953 (0.005)	100.0%	0.1%
K10 hiVar int	Fixed	0.011	0.148	0.142	0.572 (0.013)	0.939 (0.006)	100.0%	0.0%
K10 hiVar int	Ignore	0.011	0.148	0.171	0.428 (0.013)	0.972 (0.004)	100.0%	0.0%
K10 hiVar int	Random	0.011	0.148	0.142	0.573 (0.013)	0.939 (0.006)	100.0%	0.0%
K10 hiVar int imb	Fixed	-0.007	0.173	0.142	0.537 (0.013)	0.901 (0.008)	100.0%	0.0%
K10 hiVar int imb	Ignore	-0.006	0.173	0.168	0.415 (0.013)	0.934 (0.006)	100.0%	0.0%
K10 hiVar int imb	Random	-0.007	0.173	0.142	0.537 (0.013)	0.901 (0.008)	100.0%	0.2%
K10 modVar bal	Fixed	0.002	0.141	0.141	0.561 (0.013)	0.949 (0.006)	100.0%	0.0%
K10 modVar bal	Ignore	0.002	0.141	0.148	0.533 (0.013)	0.955 (0.005)	100.0%	0.0%
K10 modVar bal	Random	0.002	0.141	0.141	0.561 (0.013)	0.949 (0.006)	100.0%	3.2%
K10 modVar imbal	Fixed	-0.002	0.139	0.141	0.553 (0.013)	0.957 (0.005)	100.0%	0.0%
K10 modVar imbal	Ignore	-0.002	0.139	0.147	0.515 (0.013)	0.965 (0.005)	100.0%	0.0%
K10 modVar imbal	Random	-0.002	0.139	0.141	0.550 (0.013)	0.958 (0.005)	100.0%	8.2%
K10 modVar int	Fixed	0.004	0.156	0.142	0.559 (0.013)	0.928 (0.007)	100.0%	0.0%
K10 modVar int	Ignore	0.004	0.156	0.149	0.528 (0.013)	0.942 (0.006)	100.0%	0.0%
K10 modVar int	Random	0.004	0.156	0.142	0.559 (0.013)	0.928 (0.007)	100.0%	3.1%
K10 noVar bal	Fixed	0.003	0.140	0.142	0.569 (0.013)	0.951 (0.006)	100.0%	0.0%
K10 noVar bal	Ignore	0.003	0.140	0.142	0.568 (0.013)	0.950 (0.006)	100.0%	0.0%
K10 noVar bal	Random	0.003	0.140	0.141	0.573 (0.013)	0.950 (0.006)	100.0%	55.7%
K30 hiVar bal	Fixed	0.001	0.082	0.082	0.947 (0.006)	0.937 (0.006)	100.0%	0.0%
K30 hiVar bal	Ignore	0.001	0.082	0.099	0.898 (0.008)	0.980 (0.004)	100.0%	0.0%
K30 hiVar bal	Random	0.001	0.082	0.082	0.947 (0.006)	0.937 (0.006)	100.0%	0.0%
K30 hiVar imbal	Fixed	0.004	0.081	0.082	0.974 (0.004)	0.947 (0.006)	100.0%	0.0%
K30 hiVar imbal	Ignore	0.004	0.081	0.098	0.919 (0.007)	0.975 (0.004)	100.0%	0.0%
K30 hiVar imbal	Random	0.004	0.081	0.082	0.974 (0.004)	0.948 (0.006)	100.0%	0.0%
K30 hiVar int	Fixed	0.003	0.092	0.082	0.937 (0.006)	0.928 (0.007)	100.0%	0.0%
K30 hiVar int	Ignore	0.003	0.092	0.100	0.879 (0.008)	0.972 (0.004)	100.0%	0.0%
K30 hiVar int	Random	0.003	0.092	0.082	0.937 (0.006)	0.928 (0.007)	100.0%	0.0%
K30 hiVar int imb	Fixed	0.003	0.100	0.082	0.925 (0.007)	0.894 (0.008)	100.0%	0.0%
K30 hiVar int imb	Ignore	0.003	0.101	0.099	0.855 (0.009)	0.941 (0.006)	100.0%	0.0%
K30 hiVar int imb	Random	0.003	0.100	0.082	0.924 (0.007)	0.895 (0.008)	100.0%	0.0%
K30 modVar bal	Fixed	0.001	0.079	0.082	0.963 (0.005)	0.954 (0.005)	100.0%	0.0%
K30 modVar bal	Ignore	0.001	0.079	0.086	0.954 (0.005)	0.963 (0.005)	100.0%	0.0%
K30 modVar bal	Random	0.001	0.079	0.082	0.963 (0.005)	0.954 (0.005)	100.0%	0.1%
K30 modVar imbal	Fixed	0.000	0.081	0.082	0.959 (0.005)	0.956 (0.005)	100.0%	0.0%
K30 modVar imbal	Ignore	0.000	0.081	0.085	0.948 (0.006)	0.961 (0.005)	100.0%	0.0%
K30 modVar imbal	Random	0.000	0.081	0.082	0.956 (0.005)	0.954 (0.005)	100.0%	0.1%
K30 modVar int	Fixed	-0.002	0.091	0.082	0.936 (0.006)	0.923 (0.007)	100.0%	0.0%
K30 modVar int	Ignore	-0.002	0.091	0.086	0.918 (0.007)	0.936 (0.006)	100.0%	0.0%
K30 modVar int	Random	-0.002	0.091	0.082	0.936 (0.006)	0.923 (0.007)	100.0%	0.0%
K30 noVar bal	Fixed	0.000	0.082	0.082	0.959 (0.005)	0.943 (0.006)	100.0%	0.0%
K30 noVar bal	Ignore	0.000	0.082	0.082	0.960 (0.005)	0.943 (0.006)	100.0%	0.0%
K30 noVar bal	Random	0.000	0.082	0.081	0.960 (0.005)	0.943 (0.006)	100.0%	52.7%

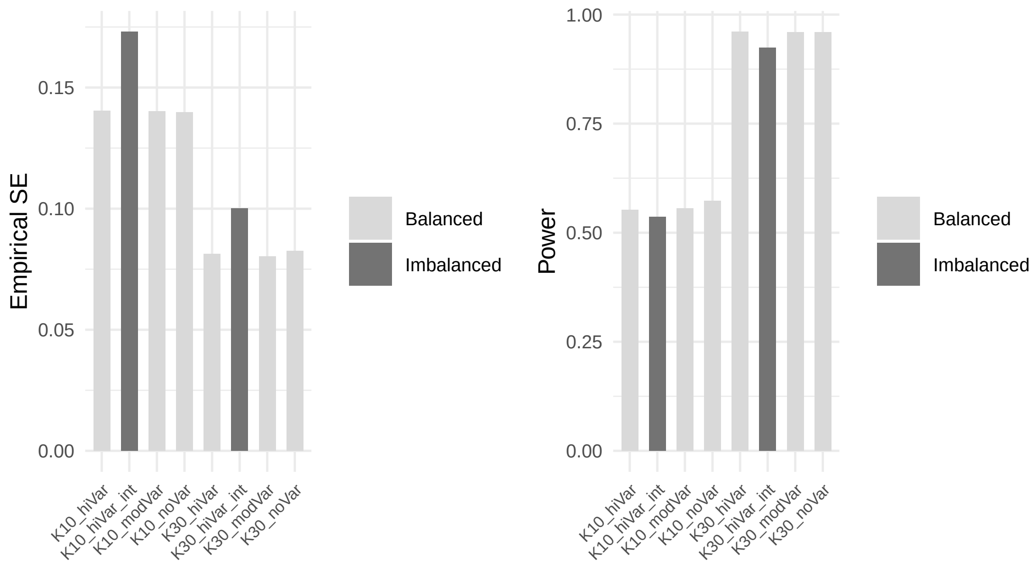

Figure 3: Comparison of balanced versus imbalanced site sizes for the random effects method. Panels show empirical standard error (left) and power (right).

3.2 Summary of performance metrics

3.3 Bias

3.4 Power and coverage

3.5 Impact of imbalanced site sizes

3.6 Type I error under the null

Coverage of a 95% CI centered on the true alternative ($\delta = 0.30$, Table 1) is not the same quantity as the Type I error rate of the associated test under a true null effect ($\delta = 0$), which is the quantity most directly relevant to the stratified-randomization literature motivating this study[11,13]. Table 2 reports the empirical rejection rate at $\alpha = 0.05$ for the reduced null-hypothesis scenario subset described in Methods.

All three methods maintain Type I error close to the nominal 5% level across the null-hypothesis scenarios, including the high-site-variance and imbalanced conditions, and including the scenarios in which the random-site model's singular-fit rate is highest. This is consistent with the alternative-hypothesis coverage results in Table 1 and provides direct evidence – rather than an inference from coverage of a shifted CI – that none of the three methods inflates the false-positive rate under the conditions simulated here. It does not, by itself, establish that the t-based reference distribution used for the random-site method (Methods, "Analytic methods") is correctly calibrated in general; the boundary/singular fits flagged in Table 1 and Table 2 return a point estimate and standard error that are numerically valid but whose sampling distribution at the parameter-space boundary is not guaranteed to match the naive t approximation used here, and this simulation's null-arm results should not be read as validating that approximation outside the specific scenarios simulated.

Table 2: Empirical Type I error under the null hypothesis ($\delta = 0$) at nominal $\alpha = 0.05$, by scenario and analytic method, with Morris et al. (2019) Table 6 Monte Carlo SE in parentheses. R = 1500 replications per scenario. 'Sing.' is the random-site singular/boundary-fit rate among converged fits.

Scenario	Method	Bias	Type I error (MCSE)	Coverage (MCSE)	Sing.
K10 hiVar bal	Fixed	0.000	0.051 (0.006)	0.949 (0.006)	0.0%
K10 hiVar bal	Ignore	0.000	0.021 (0.004)	0.979 (0.004)	0.0%
K10 hiVar bal	Random	0.000	0.051 (0.006)	0.949 (0.006)	0.1%
K10 hiVar imbal	Fixed	-0.005	0.048 (0.006)	0.952 (0.006)	0.0%
K10 hiVar imbal	Ignore	-0.005	0.019 (0.003)	0.981 (0.003)	0.0%
K10 hiVar imbal	Random	-0.005	0.046 (0.005)	0.954 (0.005)	0.2%
K10 modVar bal	Fixed	0.003	0.057 (0.006)	0.943 (0.006)	0.0%
K10 modVar bal	Ignore	0.003	0.045 (0.005)	0.955 (0.005)	0.0%
K10 modVar bal	Random	0.003	0.057 (0.006)	0.943 (0.006)	3.3%
K10 noVar bal	Fixed	0.001	0.041 (0.005)	0.959 (0.005)	0.0%
K10 noVar bal	Ignore	0.001	0.041 (0.005)	0.959 (0.005)	0.0%
K10 noVar bal	Random	0.001	0.041 (0.005)	0.959 (0.005)	56.9%
K30 hiVar bal	Fixed	0.001	0.048 (0.006)	0.952 (0.006)	0.0%
K30 hiVar bal	Ignore	0.001	0.014 (0.003)	0.986 (0.003)	0.0%
K30 hiVar bal	Random	0.001	0.048 (0.006)	0.952 (0.006)	0.0%
K30 hiVar imbal	Fixed	-0.004	0.065 (0.006)	0.935 (0.006)	0.0%
K30 hiVar imbal	Ignore	-0.004	0.023 (0.004)	0.977 (0.004)	0.0%
K30 hiVar imbal	Random	-0.004	0.068 (0.007)	0.932 (0.007)	0.0%
K30 modVar bal	Fixed	0.000	0.052 (0.006)	0.948 (0.006)	0.0%
K30 modVar bal	Ignore	0.000	0.042 (0.005)	0.958 (0.005)	0.0%
K30 modVar bal	Random	0.000	0.052 (0.006)	0.948 (0.006)	0.0%
K30 noVar bal	Fixed	0.001	0.047 (0.005)	0.953 (0.005)	0.0%
K30 noVar bal	Ignore	0.001	0.047 (0.005)	0.953 (0.005)	0.0%
K30 noVar bal	Random	0.001	0.048 (0.006)	0.952 (0.006)	53.9%

3.7 Random slopes under treatment-by-site interaction

The random-site method fitted throughout this report (Table 1, Table 2) includes only a random intercept, which is misspecified for the six treatment-by-site-interaction scenarios ($\sigma_\gamma^2 = 0.04$), since those scenarios are generated with a genuine random treatment slope, γ_i (Data generating model). Table 3 compares that intercept-only specification against the correctly specified random-slopes method (Analytic methods, Method 4), fit on the same generated data, restricted to those six scenarios.

Across these six scenarios, the random-slopes method's coverage was consistently higher than the (misspecified) random-intercept method's coverage, by +2.0 to +3.9 percentage points, and its singular-fit rate was 33.6 to 65.4%, reflecting the additional variance and covariance parameters the random-slopes model must estimate relative to the random-intercept model. See the Discussion, "Treatment-by-site interaction," for interpretation.

4 Discussion

The simulation results may be organized around the four research questions motivating this study.

Modeling site effects. Consistent with prior simulation work[7,8], we find that the random effects model attained nominal or near- nominal coverage and the highest power across nearly

Table 3: Random intercept versus random slopes for the six treatment-by-site-interaction scenarios ($\sigma_\gamma^2 = 0.04$), fit on the same generated data. 'Sing.' is the singular/boundary-fit rate among converged fits for that method; for Random slope this reflects the joint intercept-slope covariance parameter space, a higher-dimensional boundary than the single-variance-component boundary tracked for Random intercept.

Scenario	Method	Bias	Emp. SE	Power (MCSE)	Coverage (MCSE)	Sing.
K10 hiVar int	Random intercept	0.011	0.148	0.573 (0.013)	0.939 (0.006)	0.0%
K10 hiVar int	Random slope	0.011	0.148	0.456 (0.013)	0.963 (0.005)	49.7%
K10 hiVar int imb	Random intercept	-0.007	0.173	0.537 (0.013)	0.901 (0.008)	0.2%
K10 hiVar int imb	Random slope	-0.007	0.175	0.417 (0.013)	0.939 (0.006)	58.9%
K10 modVar int	Random intercept	0.004	0.156	0.559 (0.013)	0.928 (0.007)	3.1%
K10 modVar int	Random slope	0.004	0.156	0.446 (0.013)	0.956 (0.005)	65.4%
K30 hiVar int	Random intercept	0.003	0.092	0.937 (0.006)	0.928 (0.007)	0.0%
K30 hiVar int	Random slope	0.003	0.092	0.911 (0.007)	0.948 (0.006)	33.6%
K30 hiVar int imb	Random intercept	0.003	0.100	0.924 (0.007)	0.895 (0.008)	0.0%
K30 hiVar int imb	Random slope	0.003	0.101	0.873 (0.009)	0.934 (0.006)	38.0%
K30 modVar int	Random intercept	-0.002	0.091	0.936 (0.006)	0.923 (0.007)	0.0%
K30 modVar int	Random slope	-0.002	0.091	0.889 (0.008)	0.947 (0.006)	47.9%

all scenarios. The fixed effects model performed comparably when the number of sites was small ($K = 10$) but lost efficiency with $K = 30$ due to the large number of nuisance parameters. Ignoring site effects produced unbiased point estimates but inflated standard errors when site-level variance was present, leading to conservative inference and reduced power. These findings support the random effects model as a robust default for multicenter RCTs, as recommended in the literature[1,16]. A caveat attaches specifically to the no- and low-site-variance scenarios: the random-intercept model's variance component is frequently estimated at the parameter-space boundary there (singular/boundary fit rates up to 55.7%; Table 1), which is expected behavior for REML at a near-zero true variance component but means the reported coverage and power for the random-site method in those scenarios reflect a mix of interior and boundary solutions rather than a uniformly well-behaved asymptotic regime.

A methodological caveat applies to how this recommendation is justified, not to the simulation's findings themselves: the recommendation rests on a design-based argument that requires no claim about hospitals being a literal random sample of a larger population, which is the more defensible of two justifications the published literature offers for this choice; see Introduction, "Two different reasons statisticians give for treating site as random," for the full discussion and citations.

Failure to adjust for stratification. The 'ignore site' method effectively represents a failure to adjust for the stratification variable (site) used in randomization. The overcoverage and power loss observed under this method confirm the findings of Kahan and Morris[11]: when randomization is stratified by center, the analysis must account for center to obtain valid inference. The magnitude of the power penalty depends on the ICC: with no site-level variance, ignoring site is harmless, but as σ_α^2 increases, the penalty becomes substantial.

Treatment-by-site interaction. Introducing random treatment-by-site interaction ($\sigma_\gamma^2 = 0.04$) increased the variability of treatment effect estimates across replications for all methods. The random-intercept model, which does not explicitly model this interaction in the fitted specification, undercovered somewhat in Table 1 (as low as 0.895 against the 95% nominal target) because the inflated variability from the omitted slope was only partially absorbed into the residual. Table 3 (Results, "Random slopes under treatment-by-site interaction") reports what the correctly specified random-slopes model does differently on the same data: coverage was higher in all six interaction scenarios, by +2.0 to +3.9 percentage points relative to the random-intercept model, closing a real and consistent share of the coverage gap rather than a shift within Monte Carlo noise. That correction comes at a substantial cost: the random-slopes model's singular-fit rate (33.6–65.4%) was far higher than the random-intercept model's, reflecting how much harder the additional variance-covariance parameters are to estimate from a single trial's worth of data. Both facts matter for practice: a random-slopes model is not merely the theoretically correct choice under known treatment heterogeneity but delivers a measurable coverage benefit in the scenarios simulated here, while also requiring a researcher to expect and plan for a materially higher rate of boundary/singular fits than the simpler random-intercept model produces. Jones and colleagues and Fedorov and Jones have argued that some degree of treatment-by-center interaction is ubiquitous, and that the power to detect it is typically inadequate[4,15]. In our view, researchers should plan for its presence rather than test for its absence, and Table 3's results suggest that planning should weigh the random-slopes model's estimation difficulty against its real, non-trivial coverage benefit, not treat the choice as free in either direction, in a single trial's finite sample.

Imbalanced site sizes. The comparison of balanced versus imbalanced site sizes reveals that for the random effects model, moderate imbalance (CV $\approx$ 1.1) produces only a modest increase in empirical standard error and a correspondingly small decrease in power. This finding is consistent with Ruvuna and with Vierron and Giraudeau, who showed that the efficiency loss from unequal site sizes is generally small unless the imbalance is extreme[18,19]. The more concerning scenario is when site size is confounded with site characteristics – a possibility not simulated here but discussed elsewhere[15,17]. Under informative site sizes, where larger sites systematically differ from smaller sites, all methods may produce biased estimates of the population-average treatment effect.

Limitations. We should note that this simulation is restricted to continuous outcomes analyzed with normal- theory models. Binary and time-to-event outcomes raise additional complications, including separation in fixed effects logistic regression with many small sites[5,20,23]. The simulation does not incorporate dropout, non-compliance, or informative site sizes. Table 1's primary comparison of site-modeling strategies still fits only a random intercept for the random-site method, by design, since that is the question those 16 scenarios address; a random-slopes model is now fit and compared against it, but only for the six treatment-by-site-interaction scenarios (Table 3), where it is the correctly specified model. The null-hypothesis (Type I error) arm covers a reduced subset of eight of the sixteen alternative-hypothesis scenarios, omitting the treatment-by-site-interaction conditions; Type I error under a null average effect combined with genuine site-level interaction variance, analyzed by the random-slopes model, is not assessed here and remains a natural extension of the null arm. The random-site and random-slope methods' p-values and CIs both use an N-minus-fixed-effects degrees of freedom that is not a Kenward-Roger or Satterthwaite correction (Methods, "Analytic methods"); the null-arm results in Table 2 show this approximation was not anti-conservative for the random-site method in the scenarios simulated (the null arm does not cover the random-slope method, per the point above), but that is not a general guarantee, particularly for the boundary/singular fits discussed above and in Table 3.

5 Future research

Several directions warrant further investigation:

1. **Random slopes models (partially addressed).** A random treatment slope ($\delta + \gamma_i$ with γ_i estimated) is now fit and compared against the random-intercept model for the six treatment-by-site- interaction scenarios (Results, "Random slopes under treatment-by-site interaction"; Table 3). What remains is to extend the null-hypothesis arm to cover interaction scenarios analyzed by the random-slopes model, so that Type I error, not only coverage and singular-fit rate, can be assessed for that method, and to extend the K/σ^2_α/balance factorial grid used for the random-intercept comparison to the random-slopes method as well, rather than the current six-scenario subset.

2. **Non-normal outcomes.** Binary and count outcomes require generalized linear mixed models or GEE, and the relative performance of methods may differ from the continuous case, particularly with sparse data per site[20,23].

3. **Informative site sizes.** If site size is associated with site-level treatment effect (e.g., expert centers enroll more patients and also deliver more effective treatment), standard methods may yield biased population-average estimates. Inverse-probability weighting by site or joint modeling of enrollment and outcome merit investigation[17].

4. **Small-sample corrections.** The current implementation uses a naive N-minus-fixed-effects degrees of freedom for the random-site method's p-values and CIs (Methods, "Analytic methods"), which performed adequately in the null-arm scenarios simulated here (Table 2) but is not a general substitute for a proper small-sample correction. Kenward-Roger or Satterthwaite denominator degrees of freedom corrections (available in `lmerTest`) should be substituted and evaluated across the same scenario grid, particularly for the $K = 10$, high- site-variance conditions where few sites most threaten the adequacy of any df approximation.

5. **Bayesian approaches.** Hierarchical Bayesian models offer partial pooling of site-specific treatment effects, which may outperform frequentist random effects models when the number of sites is very small[4].

6. **Sample size planning.** Tools that incorporate the design effect from multicenter structure, including anticipated ICC and site size imbalance, into prospective sample size calculations would improve trial planning[19,24].

6 Data and code availability

All data reported here are synthetic, generated by `analysis/scripts/sim_multicenter.R`; no external or human-participant data are used. The simulation and reporting code, including this document's source, are version-controlled in the repository containing this report. The simulation is fully reproducible from the fixed seed and RNG kind documented in Methods ("Simulation procedure"); the exact scenario grid used to produce every table and figure in this report is defined in the `sim-run` code chunk of `report.Rmd`. A fixed seed and RNG kind alone do not guarantee identical numeric results across package versions, since `lme4`'s optimizer and convergence-warning behavior (directly relevant to the singular/boundary-fit diagnostics reported here) can change between releases. The exact computational environment, including the R and `lme4` versions used to produce this report, is pinned in `renv.lock` at the repository root; running `renv::restore()` from the repository root recreates that environment exactly.

7 Conflict of interest

The author declares no conflict of interest. No external funding supported this work.

8 References

8.1 Morris et al. (2019) ADEMP Compliance

We audited this simulation study against the reporting standards proposed by Morris, White, and Crowther[21]. The original audit is recorded at `docs/morris-audit-2026-04-17.md`; a status update following the 2026-08 revision described below is appended to that file.

Verdict: Partially compliant (updated).

Resolved since the original 2026-04-17 audit:

- Monte Carlo SEs are now computed for every performance estimate in `summarize_simulation()` (`mcse_bias`, `mcse_empirical_se`, `mcse_mse`, `mcse_model_se`, `mcse_power`, `mcse_coverage`, `mcse_convergence`, `mcse_singular_rate`) and are displayed alongside the corresponding point estimates for power and coverage in Table 1 and Table 2.
- `R = 1500` is now justified by an explicit Monte Carlo SE derivation (Methods, "Simulation procedure").
- Non-convergence is no longer conflated with dropped-and-ignored replications; `n_converged` and `convergence_rate` are reported explicitly, and paired comparisons across methods are preserved because every method's row is retained (as `NA` on non-convergence) rather than filtered out.
- `RNGkind("L'Ecuyer-CMRG")` is pinned immediately before the single `set.seed()` call.
- Per-replicate `.Random.seed` values are captured by `run_simulation()`, attached as the `rng_states` attribute of its return value, and persisted to `analysis/data/derived_data/rng_states.rds` (keyed by scenario label and replicate index), so a specific failing or singular replication from a completed run can be re-run in isolation via `with_rng_state()`-style `.Random.seed` restoration without re-executing the whole simulation up to that point.

- A random-slopes method (Method 4 in "Analytic methods") is now implemented in `sim_multicenter.R::fit_methods()`, fit on every replicate, and reported in a dedicated Results subsection ("Random slopes under treatment-by-site interaction") and Table 3 for the six treatment-by-site-interaction scenarios, with the coverage gain and singular-fit rate reported in the abstract.

Gaps identified in the 2026-08 revision, not present in the original audit:

- Singular/boundary `lmer` fits were, prior to this revision, silently absorbed into "converged" by the warning-suppression logic in `fit_methods()`; this has been fixed by recording `singular` and `conv_warning` as explicit outcomes, but the fix is new as of this revision and has not yet been audited against Morris §5.1 on a wider scenario set than the one simulated here.
- Coverage previously used a fixed $z = 1.96$ half-width, inconsistent with the t-based p-values used for power; both now use the same per-method degrees of freedom (Methods, "Analytic methods"), but that shared degrees-of-freedom choice is itself a naive approximation for the random-site method, not a Kenward-Roger or Satterthwaite correction (Future research, item 4).
- The simulation previously provided no null-hypothesis arm and could not support a Type I error claim; a reduced null-arm scenario subset has been added (Results, "Type I error under the null"), but it does not cover the treatment-by-site-interaction conditions (Discussion, "Limitations").

Still open:

- The random-slopes method (Method 4) is reported only for the six treatment-by-site-interaction scenarios under the true alternative; the null-hypothesis arm does not yet cover the random-slopes method, and the K/σ_α^2/balance factorial grid used for the random-intercept comparison has not been extended to the random-slopes method (Future research, item 1; Discussion, "Limitations").

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